

Radiation - Geiger Counter Phys 20 CL Lab

Ben Leon

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1 Abstract

In this experiment, the count rates from $^{90}_{38}\text{Sr}$ and $^{60}_{27}\text{Co}$ radioactive sources were measured using a Geiger counter. The rates were calculated by counting the number of detected decays over a measured time interval, with uncertainties determined using Poisson counting statistics. Measurements were taken with the sources placed in different orientations and distances from the detector, and absorber materials were used to compare how beta and gamma radiation interact with matter. Overall, the strontium source showed a much stronger dependence on orientation and shielding, while the cobalt source was less affected by the aluminum absorber, consistent with the more penetrating nature of gamma radiation.

2 Procedure & Theory

2.1 Experimental Uncertainty

Firstly, a discussion of the methods in handling uncertainty in the experiments is necessary. The main goal is to measure the decay rate, and the rate for each trial is measured by simply dividing the number of counts detected by the time elapsed since pressing the count button:

$$R = \frac{n}{T} \quad (1)$$

The error for each run is given by the Poisson uncertainty in the rate:

$$\sigma_R = \frac{\sqrt{n}}{T} \quad (2)$$

so that our measurements of the rate take the form:

$$R \pm \sigma_R \quad (3)$$

To lower the error on the results, it is useful to perform repeated trials and attempt to get a similar number of counts each time, then to use the sum of the

values as a better measurement of the rate:

$$R_{\text{sum}} = \frac{\sum n_i}{\sum T_i} \quad (4)$$

$$\sigma_{\text{sum}} = \frac{\sqrt{\sum n_i}}{\sum T_i} \quad (5)$$

The mean of the data was also taken for experiments with repeated trials along with the standard deviation using spreadsheets:

$$\bar{R} = \frac{1}{N} \sum R_i \quad (6)$$

$$s_R = \sqrt{\frac{1}{N-1} \sum (R_i - \bar{R})^2} \quad (7)$$

2.1.1 Solid Angle Adjustment

The detecting instrument, the Geiger tube, was constructed in such a way that the decays penetrating the face of the detector are just a portion of solid angle because it would be difficult to completely surround the source with a detector. This adjustment is done after calculating the rate of decays and the error in that rate because those are generally independent of the solid angle that the detector subtends. Figure 1 shows how not all of the rays emitted by the source are actually detected, as most of them do not move towards the thin window. The fraction of solid angle that the detector subtends is given by

$$f = \frac{1}{2} (1 - \cos \theta) \quad (8)$$

where θ is the half-angle of the cone that defines the solid angle. Further measuring this fraction, we can determine the $\cos \theta$ term by using the relationship between the distance from the source to the detector and its radius.

$$\cos \theta = \frac{L}{\sqrt{L^2 + a^2}} \quad (9)$$

where a is the radius of the thin window, and L is the source-to-window distance.

To implement this correction, $R \pm \sigma_R$ is divided by f .

2.2 Background Rate

The experiments began with a measurement of the background counting rate. The purpose of measuring this is due to the ^{40}K radiation that resides primarily in the walls of Broida Hall. This can influence the measurements when making counts during the experiment. Some of the experiments involve lead absorbers being placed above the radioactive sources and will require subtracting this background rate. I got a measurement of the background rate of:

$$R_b = .34 \pm .04 \text{ cps} \quad (10)$$

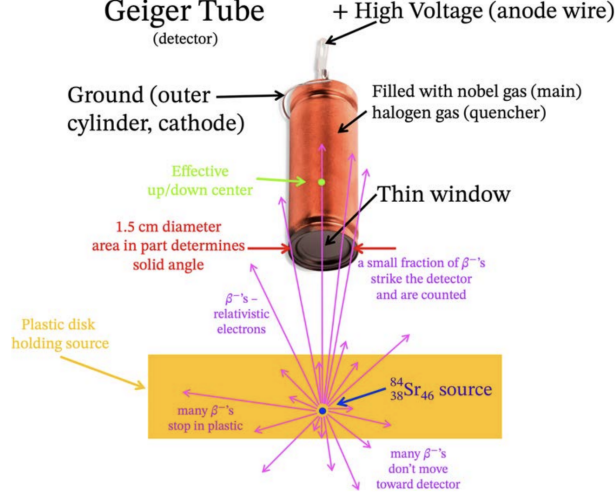


Figure 1: A representation of β^- being emitted from the source, some of them not energetic enough to leave the plastic. The depiction of the source being closer to one side also demonstrates the reasoning in doing experiments with the label faced both up and down, as more rays escape on the side closer to the edge of the plastic. The partial geometry corrects for the thin window not being able to detect the rays that do not penetrate it, but still escape the plastic nonetheless.

3 Data & Analysis

Configuration	Source	Counts, n	Time, T (s)	$R = \frac{n}{T} \pm \sigma_R$ (cps)	$\bar{R}$ (cps)
Slot 2 face up	Sr	4355	141	30.89 ± 0.47	30.94 ± 0.46
Slot 2 face up	Co	4979	553	9.00 ± 0.13	9.02 ± 0.10
Slot 2 face down	Sr	5291	52	101.75 ± 1.40	101.86 ± 1.33
Slot 2 face down	Co	5161	566	9.12 ± 0.13	9.13 ± 0.10
Slot 6 face up	Sr	1045	223	4.69 ± 0.14	—
Slot 6 face up	Co	986	465	2.12 ± 0.07	—
Slot 6 face down	Sr	1005	76	13.22 ± 0.42	—
Slot 6 face down	Co	949	486	1.95 ± 0.06	—
170 mg/cm ² absorber	Sr	3078	62	49.65 ± 0.89	—
7367 mg/cm ² absorber	Sr	100	314	0.32 ± 0.03	—
170 mg/cm ² absorber	Co	706	82	8.61 ± 0.32	—
7367 mg/cm ² absorber	Co	504	88	5.73 ± 0.26	—

Table 1: Measured count rates for strontium and cobalt sources in different slot configurations. The rate is calculated using $R = n/T$, with uncertainty $\sigma_R = \sqrt{n}/T$. Rates and errors shown are calculated from summed count measurements, lowering the overall error. Slot 6 data and data taken with absorbers only had 1 trial, so there was no mean to be taken.

The measured count rates for the strontium and cobalt sources are shown in Table 1. The data generally show that the strontium source produced much larger count rates than the cobalt source, especially when the source was placed close to the detector. This makes sense because the strontium source emits β^- particles that are counted efficiently when they reach the thin window of the Geiger tube. The cobalt source emits more penetrating γ radiation, but the Geiger tube does not detect these photons as directly or efficiently, so the measured count rates are lower.

For the slot 2 measurements, repeated trials were taken and then summed in order to reduce the statistical uncertainty. Since the Poisson error scales as $\sqrt{n}$, taking more counts lowers the fractional uncertainty in the final rate. This is why the summed measurements give a better estimate of the count rate than any individual run by itself. The mean rate $\bar{R}$ was also calculated from the trial rates, and the standard deviation of the trial rates gives a way to check how consistent the repeated measurements were.

One of the clearest results is the difference between the face-up and face-down measurements for strontium. In slot 2, the strontium face-up rate was

$$R_{\text{Sr, up}} = 30.89 \pm 0.47 \text{ cps},$$

while the face-down rate was

$$R_{\text{Sr, down}} = 101.75 \pm 1.40 \text{ cps}.$$

The face-down rate was much larger, which suggests that the radioactive material is closer to one side of the plastic disk. When that side faces the detector, more β^- particles are able to escape the plastic and reach the Geiger tube. This effect was much less significant for cobalt, where the slot 2 face-up and face-down rates were

$$R_{\text{Co, up}} = 9.00 \pm 0.13 \text{ cps}$$

and

$$R_{\text{Co, down}} = 9.12 \pm 0.13 \text{ cps}.$$

These are essentially the same within uncertainty, which is reasonable because γ rays are much more penetrating and are not as strongly affected by which side of the plastic source faces the detector.

The distance from the detector also had a clear effect on the measured rates. For strontium, moving from slot 2 to slot 6 lowered the face-up rate from

$$30.89 \pm 0.47 \text{ cps}$$

to

$$4.69 \pm 0.14 \text{ cps}.$$

This decrease is expected because the detector subtends a smaller solid angle when the source is farther away. In other words, fewer of the emitted particles are moving in a direction that allows them to enter the thin window of the

Geiger tube. The same general trend was seen for cobalt, although the rates were lower overall.

Using the solid angle fractions $f = 0.01319184472$ for slot 2 and $f = 0.002690795555$ for slot 6, the measured rates can be corrected to estimate the rate into all directions:

$$R_{4\pi} = \frac{R}{f}. \quad (11)$$

For strontium, the slot 2 face-up corrected rate was

$$R_{4\pi} = \frac{30.89}{0.01319184472} = 2342 \pm 36 \text{ cps},$$

while the slot 6 face-up corrected rate was

$$R_{4\pi} = \frac{4.69}{0.002690795555} = 1743 \pm 52 \text{ cps}.$$

These corrected rates are closer than the raw measured rates, which shows why the solid angle correction matters. They are not exactly the same, likely because the source is not a perfect point source, the plastic casing absorbs some beta particles, and the detector geometry is only being approximated.

For the face-down strontium measurements, the corrected rates were

$$R_{4\pi} = 7713 \pm 106 \text{ cps}$$

for slot 2 and

$$R_{4\pi} = 4913 \pm 156 \text{ cps}$$

for slot 6. Again, the corrected values are closer than the raw rates, but still not identical. This supports the idea that geometry explains a large part of the distance dependence, but the source orientation and absorption in the plastic disk also matter.

To compare the slot 2 and slot 6 corrected rates more directly, the difference was calculated in units of the combined uncertainty:

$$N_\sigma = \frac{|R_{4\pi,2} - R_{4\pi,6}|}{\sqrt{\sigma_2^2 + \sigma_6^2}}. \quad (12)$$

For strontium face-up, this gives

$$N_\sigma = \frac{|2342 - 1743|}{\sqrt{36^2 + 52^2}} = 9.5,$$

and for strontium face-down,

$$N_\sigma = \frac{|7713 - 4913|}{\sqrt{106^2 + 156^2}} = 14.8.$$

These differences are much larger than the statistical uncertainty alone, meaning that the disagreement between slot 2 and slot 6 is not just random counting

error. It is likely dominated by systematic effects, such as the source not acting like a perfect point source, the source material being embedded unevenly in the plastic disk, and the solid-angle correction being only an approximation.

The corrected rates can also be compared with the expected activity from the pre-lab calculation. The sources were made in February 2017, so by May 2026 the elapsed time was about 9.25 years. The activity at time t is

$$A(t) = A_0 \left(\frac{1}{2} \right)^{t/\tau_{1/2}}. \quad (13)$$

For the ^{90}Sr source, using $A_0 = 0.1 \mu\text{Ci}$ and $\tau_{1/2} = 28.91 \text{ yr}$ gives

$$A_{\text{Sr}} = 0.0801 \mu\text{Ci}.$$

Since $1 \mu\text{Ci} = 3.7 \times 10^4 \text{ Bq}$,

$$A_{\text{Sr}} = 2.96 \times 10^3 \text{ Bq}.$$

Because ^{90}Sr is in equilibrium with its ^{90}Y daughter, two β^- particles are emitted per parent decay. Therefore, the expected beta emission rate is

$$R_{\beta, \text{expected}} = 2A_{\text{Sr}} = 5.93 \times 10^3 \text{ cps}.$$

The closest corrected measurement was the slot 2 face-down value,

$$R_{4\pi} = 7713 \pm 106 \text{ cps}.$$

This is the same order of magnitude as the expected beta emission rate, but it is still noticeably higher. The difference is most likely due to systematic uncertainty in the geometric correction and the fact that the real source and detector setup are not ideal.

For the ^{60}Co source, using $A_0 = 1.0 \mu\text{Ci}$ and $\tau_{1/2} = 5.271 \text{ yr}$ gives

$$A_{\text{Co}} = 0.296 \mu\text{Ci} = 1.10 \times 10^4 \text{ Bq}.$$

Since Co-60 usually emits two gamma rays per decay, the expected gamma emission rate is roughly

$$R_{\gamma, \text{expected}} \approx 2A_{\text{Co}} = 2.19 \times 10^4 \text{ cps}.$$

The slot 2 face-up corrected cobalt rate was

$$R_{4\pi} = \frac{9.00}{0.01319184472} = 682 \pm 10 \text{ cps},$$

and the slot 2 face-down corrected cobalt rate was

$$R_{4\pi} = \frac{9.12}{0.01319184472} = 691 \pm 10 \text{ cps}.$$

These are only about 3.1% of the expected gamma emission rate. This makes sense because the Geiger tube does not efficiently detect gamma rays directly. Instead, the gamma rays mostly have to scatter electrons into the tube, which is a low-efficiency process.

The absorber measurements show the difference between how β^- particles and γ rays interact with matter. For strontium, the rate with the 170 mg/cm² aluminum absorber was

$$49.65 \pm 0.89 \text{ cps},$$

while with the 7367 mg/cm² lead absorber it was reduced to

$$0.32 \pm 0.03 \text{ cps}.$$

This lead absorber rate is extremely close to the measured background rate, meaning that most of the strontium beta radiation was stopped by the absorber.

For cobalt, the aluminum absorber gave a rate of

$$8.61 \pm 0.32 \text{ cps},$$

while the lead absorber gave

$$5.73 \pm 0.26 \text{ cps}.$$

The aluminum absorber did not reduce the cobalt rate very much, but the lead absorber had a more noticeable effect. This is consistent with cobalt radiation being mostly γ radiation, which is more penetrating than beta radiation and therefore requires denser shielding to significantly reduce the count rate.

The absorber ratios can also be used to compare the strength of attenuation:

$$\rho = \frac{R_{\text{absorber}}}{R_{\text{no absorber}}}. \quad (14)$$

A value of ρ close to 1 means the absorber did not have much effect, while a value much less than 1 means the absorber strongly reduced the radiation. Using the slot 2 face-down measurements as the no-absorber comparison, the absorber ratios were calculated for each source.

For strontium, the aluminum absorber gave

$$\rho_{\text{Sr, Al}} = \frac{49.65}{101.75} = 0.488 \pm 0.011,$$

while the lead absorber gave

$$\rho_{\text{Sr, Pb}} = \frac{0.32}{101.75} = 0.0031 \pm 0.0003.$$

This means the aluminum absorber reduced the strontium count rate by about half, while the lead absorber reduced it almost completely to the background level.

For cobalt, the aluminum absorber gave

$$\rho_{\text{Co, Al}} = \frac{8.61}{9.12} = 0.944 \pm 0.038,$$

while the lead absorber gave

$$\rho_{\text{Co, Pb}} = \frac{5.73}{9.12} = 0.628 \pm 0.029.$$

This shows that cobalt radiation was barely reduced by aluminum but was reduced more noticeably by lead. This agrees with the expectation that gamma radiation is more penetrating than beta radiation.

4 Conclusion

In this lab, the count rates of $^{90}_{38}\text{Sr}$ and $^{60}_{27}\text{Co}$ sources were measured with a Geiger counter. The uncertainty in each rate was found using Poisson statistics, where the error in the number of counts is proportional to $\sqrt{n}$. Repeated trials and summed counts were useful because they reduced the fractional uncertainty in the final measured rates.

The strontium source showed a strong dependence on orientation, with the face-down orientation producing a much larger count rate than the face-up orientation. This suggests that the radioactive material inside the plastic disk is closer to one side, allowing more beta particles to escape when that side faces the detector. The cobalt source did not show as much dependence on orientation, which is expected because gamma rays are more penetrating.

The absorber data showed that beta radiation from strontium was strongly reduced by shielding, especially by the lead absorber, which brought the count rate close to the background level. Cobalt radiation was less affected by the aluminum absorber and only partially reduced by lead, which is consistent with gamma radiation being more penetrating than beta radiation. Overall, the measurements agreed with the expected behavior of beta and gamma radiation and demonstrated the importance of counting statistics, geometry, and absorption in radiation measurements.